Homework 3 – Regression

1 Question 1

Below, I simulate a rather quite annoying set of regression data.

```
set.seed(0510)
n <- 250
dat.q1 <- tibble(x1 = rnorm(n, mean = 50, sd = 10),
                 x2 = factor(sample(c("A", "B"), n, replace = TRUE)),
                 x3 = factor(sample(c("Control", "Treat"), n, replace = TRUE)),
                 x4 = factor(sample(c("Low", "High"), n, replace = TRUE), levels=c("Low", "High")),
                 x5 = rnorm(n, mean = 0, sd = 1),
                 x6 = factor(sample(c("Group1", "Group2"), n, replace = TRUE)),
                 x7 = rnorm(n, mean = 10, sd = 2),
                 x8 = rnorm(n, mean = 10, sd = 2)) |>
        mutate(y = 5 +                          # Intercept
                   # Quantiative x1
                   2 * x1 +
                   # Categorical x2
                   (x2 == "B") * 10 +
                   # Categorical x Categorical (x3*x4)
                   (x3 == "Treat") * 5 + (x4 == "High") * 2 +
                   (x3 == "Treat" & x4 == "High") * 15 +
                   # Quantitative x Categorical  (x5*x6)
                   3 * x5 + (x6 == "Group2") * 4 +
                   (x5 * (x6 == "Group2")) * 8 +
                   # Quantitative x Quantitative (x7*x8)
                   1.5 * x7 + 0.5 * x8 +
                   0.8 * (x7 * x8) +
                   # Error
                   rnorm(n, mean = 0, sd = 15))
```

1.1 Part a

Fit the regression model.

$$
\begin{aligned}
Y_i = &\beta_0 + \beta_1 x_{1i} + \beta_2 I(x_{2i} = \text{B}) + \beta_3 I(x_{3i} = \text{Treat}) + \beta_4 I(x_{4i} = \text{High}) + \\
&\beta_5 x_{5i} + \beta_6 I(x_{6i} = \text{Group2}) + \beta_7 x_{7i} + \beta_8 x_{8i} + \\
&\beta_9 (I(x_{3i} = \text{Treat}) \times I(x_{4i} = \text{High})) + && \text{(Categorical} \times \text{Categorical)} \\
&\beta_{10} (x_{5i} \times I(x_{6i} = \text{Group2})) + && \text{(Continuous} \times \text{Categorical)} \\
&\beta_{11} (x_{7i} \times x_{8i}) + \epsilon_i && \text{(Continuous} \times \text{Continuous)}
\end{aligned}
$$

Solution: First, we fit the regression model using `lm()`.

```
mod.q1 <- lm(y ~ x1 + x2 + x3*x4 + x5*x6 + x7*x8, data=dat.q1)
```

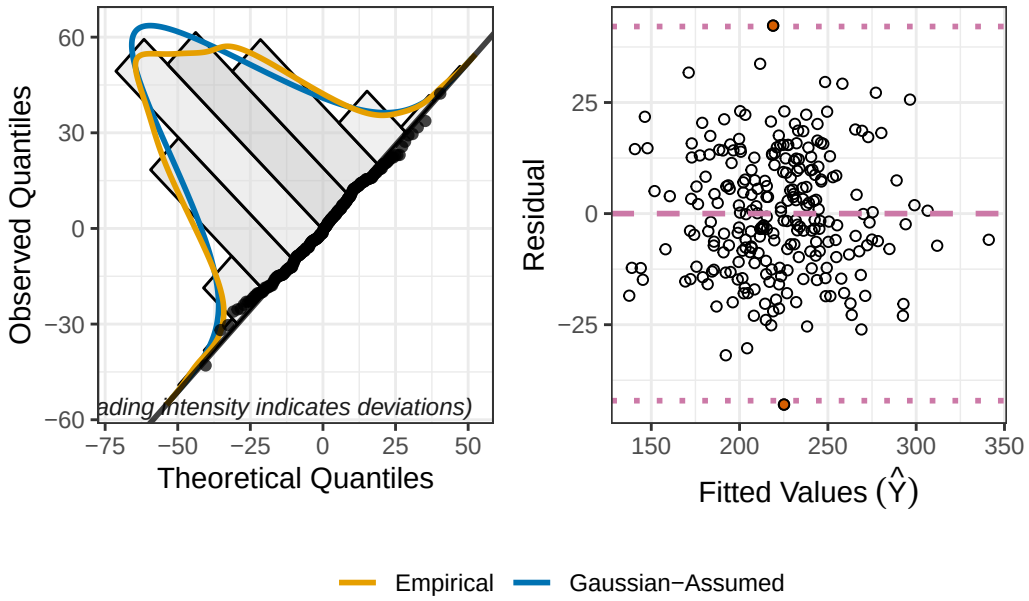
1.2 Part b

Conduct a residual and outlier analysis.

Solution

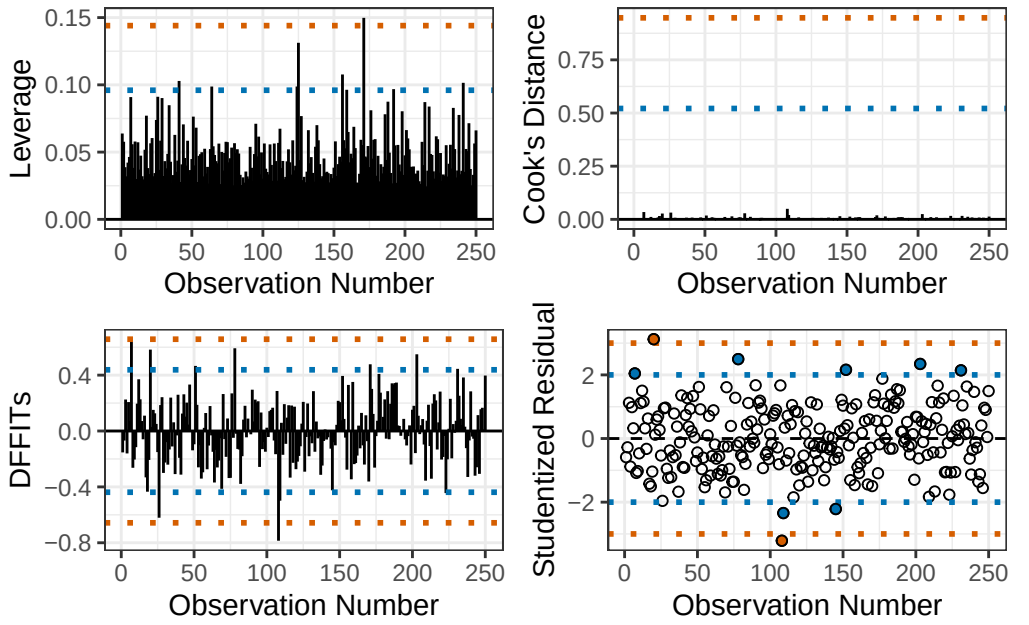
```
plotResiduals(mod.q1)
```

Figure 1: Residual Analysis of Simulated Data




```
plotInfluence(mod.q1)
```

Figure 2: Outlier Analysis of Simulated Data



As shown in the residual scott plot in Figure 1, the residuals appear uniformly distributed around 0, with both a highly positive and negative outlier preent. The distribution of the residuals in the quantile plot indicates approximate normality. With respect to the outliers, as shown in Figure 2, while there are a few high leverage points and unusual points, the cooks distance and DFFITs plots suggest that they are not problematic for the regression.

1.3 Part c

Interpret the intercept, the coefficients on x_1 and x_2.

Solution

The intercept β_0, is the expectation of Y in the absence of any other continuous or categorical features of the model. Thus it's what the model expects of Y if all continuous variables (x_1, x_5, x_7, x_8) are 0, and all of the categorical variables (x_2, x_4, x_6) are 0, meaning that the observation is not in any of the categories. The coefficient β_1 indicates the increase in the expectation of Y for a per unit increase in x_1, while the coefficient β_2 is the increase in expectation of Y due to membership in category B.

1.4 Part d

Untangle and interpret the $x_3 * x_4$ interaction using algebra. Use Table 1 and Figure 3 for context.

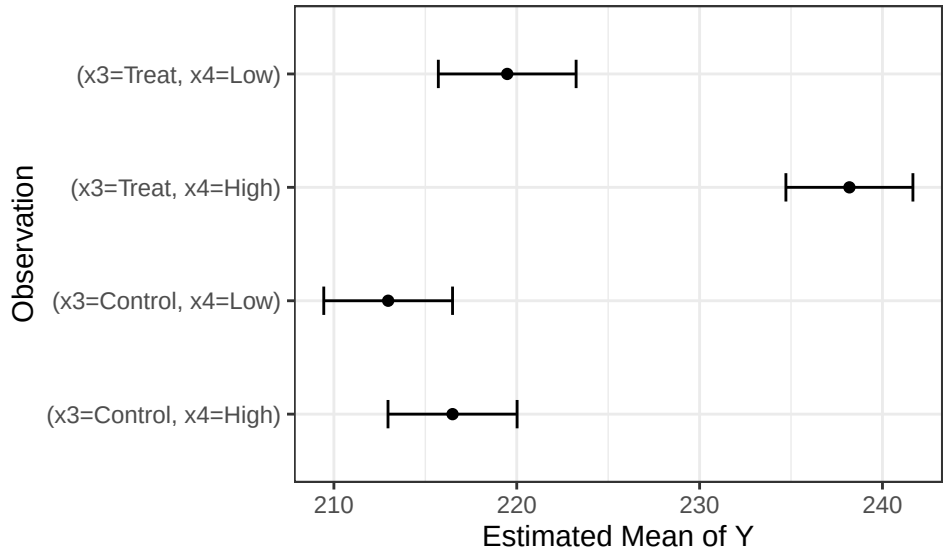
```
library(emmeans)
emm.q1d <- emmeans(mod.q1, ~x3*x4) |> # Don't average across levels: use |x2*x6
            data.frame() |>
            rename("Mean"  = "emmean",
                   "Lower" = "lower.CL",
                   "Upper" = "upper.CL")
```

Table 1: Estimated marginal means for the model in Question 1.

x3	x4	Mean	SE	df	Lower	Upper
Control	Low	212.9676	1.7871	238	209.4470	216.4882
Treat	Low	219.4792	1.9123	238	215.7120	223.2464
Control	High	216.4894	1.7916	238	212.9601	220.0188
Treat	High	238.1930	1.7631	238	234.7198	241.6662

```
emm.q1d <- emm.q1d |>
  mutate(Observation = paste("(x3=", x3, ", ", "x4=", x4, ")", sep=""))
ggplot(emm.q1d) +
  geom_point(aes(x=Mean, y=Observation))+
  geom_errorbar(aes(y=Observation,
                    xmin=Lower, xmax=Upper),
                width = 0.25)+
  # facet_wrap(~x2*x6)+
  theme_bw()+
  xlab("Estimated Mean of Y")
```

Figure 3: Estimated marginal means for the model in Question 1.



Solution Given the interaction $x_3 \times x_4$, given this is a categorical x categorical interaction, we consider the state of the model under all combinations of values for $I(x_{3i} == Treat), I(x_{4i} == High)$

- **(Control, low)**

$$\begin{aligned} Y_i =& \beta_0 + \beta_1 x_{1i} + \beta_2 I(x_{2i} = \text{B}) + \\ & \beta_5 x_{5i} + \beta_6 I(x_{6i} = \text{Group2}) + \beta_7 x_{7i} + \beta_8 x_{8i} + \\ & \beta_{10}(x_{5i} \times I(x_{6i} = \text{Group2})) \\ & \beta_{11}(x_{7i} \times x_{8i}) + \epsilon_i \end{aligned}$$

- **(Treat, low)**

$$\begin{aligned} Y_i =& \beta_0 + \beta_1 x_{1i} + \beta_2 I(x_{2i} = \text{B}) + \\ & \beta_3 + \\ & \beta_5 x_{5i} + \beta_6 I(x_{6i} = \text{Group2}) + \beta_7 x_{7i} + \beta_8 x_{8i} + \\ & \beta_{10}(x_{5i} \times I(x_{6i} = \text{Group2})) + \\ & \beta_{11}(x_{7i} \times x_{8i}) + \epsilon_i \end{aligned}$$

- **(Control, High)**

$$\begin{aligned} Y_i =& \beta_0 + \beta_1 x_{1i} + \beta_2 I(x_{2i} = \text{B}) + \\ & \beta_4 + \\ & \beta_5 x_{5i} + \beta_6 I(x_{6i} = \text{Group2}) + \beta_7 x_{7i} + \beta_8 x_{8i} + \\ & \beta_{10}(x_{5i} \times I(x_{6i} = \text{Group2})) + \\ & \beta_{11}(x_{7i} \times x_{8i}) + \epsilon_i \end{aligned}$$

- **(Treat, High)**

$$\begin{aligned} Y_i =& \beta_0 + \beta_1 x_{1i} + \beta_2 I(x_{2i} = \text{B}) + \\ & \beta_3 + \beta_4 + \\ & \beta_5 x_{5i} + \beta_6 I(x_{6i} = \text{Group2}) + \beta_7 x_{7i} + \beta_8 x_{8i} + \\ & \beta_9 + \\ & \beta_{10}(x_{5i} \times I(x_{6i} = \text{Group2})) + \\ & \beta_{11}(x_{7i} \times x_{8i}) + \epsilon_i \end{aligned}$$

Using the values in the table:

- $\beta_3 = 219.4792 - 212.9676 = 6.5116$
- $\beta_4 = 216.4894 - 212.9676 = 3.5218$
- $\beta_9 = 238.1930 - 212.9676 - 3.5218 - 6.5116 = 15.192$ Thus, we interpret the total effect of x_3, x_4 to be 25.2254, 6.5116 due to membership in the treatment group, 3.5218 due to membership in the high group, and 15.192 the marginal increase due to membership in both groups.

1.5 Part e

Untangle and interpret the $x_5 * x_6$ interaction using algebra. Use Table 2 and Figure 4 for context.

```
emm.q1e <- emtrends(mod.q1, var="x5", ~x5|x6) |>
           data.frame() |>
           rename("Slope"  = "x5.trend",
                  "Lower" = "lower.CL",
                  "Upper" = "upper.CL") |>
           dplyr::select(x6, Slope, SE, df, Lower, Upper)
```

Table 2: Estimated marginal trends for the model in Question 1.

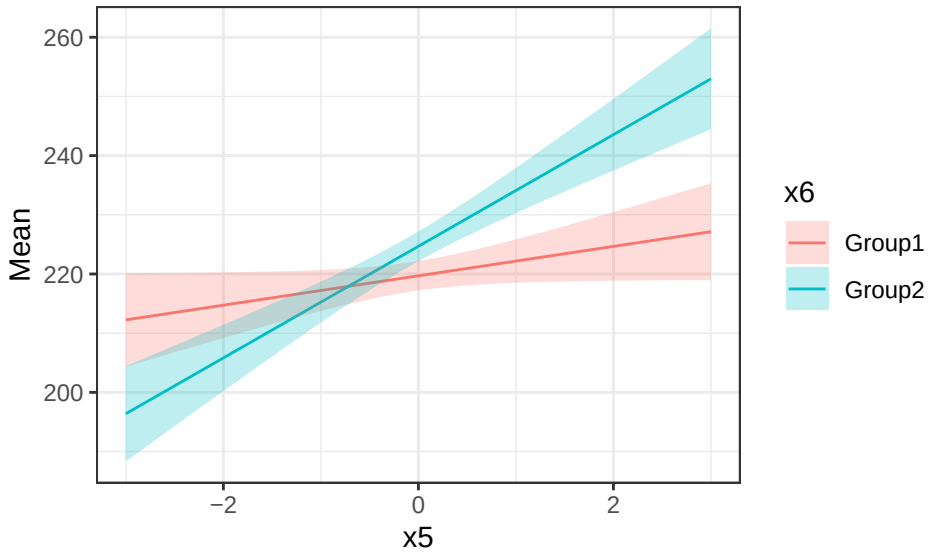
x6	Slope	SE	df	Lower	Upper
Group1	2.4835	1.2942	238	-0.0660	5.0330
Group2	9.4309	1.3367	238	6.7977	12.0642

In Figure 4, the slopes are the same and just the intercepts vary depending on x_2, x_3, and x_4.

```
emm.q1e <- emmeans(mod.q1, ~x5*x6,
                   at = list(x5 = seq(-3, 3, length.out=500))) |> # Don't average across levels: use |x2*x3*x4
  data.frame() |>
            rename("Mean"  = "emmean",
                   "Lower" = "lower.CL",
                   "Upper" = "upper.CL")

ggplot(emm.q1e) +
  geom_line(aes(x=x5, y=Mean, color=x6))+
  geom_ribbon(aes(x=x5, ymin=Lower, ymax=Upper, fill=x6), alpha=0.25)+
  # facet_wrap(~x2*x3*x4)+
  theme_bw()
```

Figure 4: Estimated marginal trends for the model in Question 1.



Solution As shown in Figure 4, the interaction coefficient β_{10} refers to the the slope of the group 2 line, 9.4309, while the coefficient β_5 refers to the slope of the group 1 line. The coefficient β_6 is the increase in expected Y due to membership in group 2, and thus the difference in y-intercept between the group 1 line and the group 2 line.

1.6 Part f

Untangle and interpret the $x_7 * x_8$ interaction using algebra. Use Table 3 and Table 4 for context.

```
emm.q1f.x7 <- emtrends(mod.q1, var="x7", ~x7|x8,
                   at = list(x8 = mean(dat.q1$x8) + c(-1, 0 ,1)*sd(dat.q1$x8))) |>
            data.frame() |>
            rename("Slope"  = "x7.trend",
                   "Lower" = "lower.CL",
                   "Upper" = "upper.CL") |>
            dplyr::select(x8, Slope, SE, df, Lower, Upper)
```

Table 3: Estimated marginal trends for the model in Question 1.

x8	Slope	SE	df	Lower	Upper
8.1309	7.9230	0.6751	238	6.5932	9.2529

Table 3: Estimated marginal trends for the model in Question 1.

x8	Slope	SE	df	Lower	Upper
10.0582	9.7129	0.5015	238	8.7250	10.7008
11.9854	11.5027	0.7396	238	10.0458	12.9597

```r
emm.q1f.x8 <- emtrends(mod.q1, var="x8", ~x8|x7,
                       at = list(x7 = mean(dat.q1$x7) + c(-1, 0 ,1)*sd(dat.q1$x7))) |>
              data.frame() |>
              rename("Slope"  = "x8.trend",
                     "Lower" = "lower.CL",
                     "Upper" = "upper.CL") |>
              dplyr::select(x7, Slope, SE, df, Lower, Upper)
```

Table 4: Estimated marginal trends for the model in Question 1.

x7	Slope	SE	df	Lower	Upper
8.3824	6.8569	0.6069	238	5.6613	8.0525
10.2208	8.5643	0.4744	238	7.6297	9.4989
12.0592	10.2716	0.7325	238	8.8286	11.7146

A Johnson Neyman analysis using the `interactions` package for `R` (Long 2024), shows that the slope of x_7 is statistically discernible for all observed values of x_8 (Figure 5), and vice versa (Figure 6).

```r
library(interactions)
johnson_neyman(mod.q1, pred = x7, modx = x8, control.fdr = T)
```

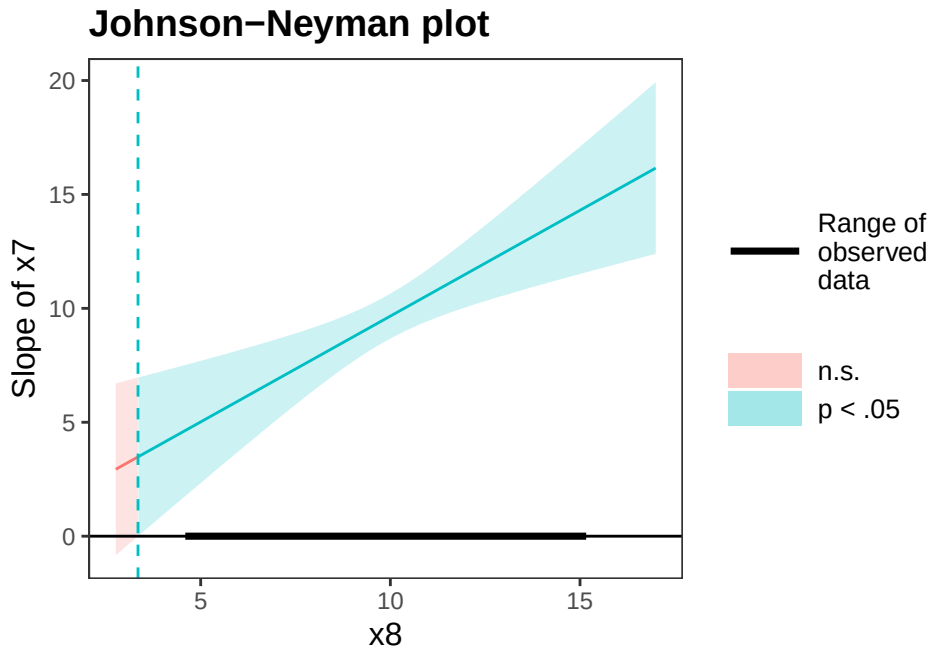
```
JOHNSON-NEYMAN INTERVAL

When x8 is OUTSIDE the interval [-13.08, 3.35], the slope of x7 is p < .05.

Note: The range of observed values of x8 is [4.69, 15.07]

Interval calculated using false discovery rate adjusted t = 1.97
```

Figure 5: Johnson Neyman interval for the model in Question 1.




```r
johnson_neyman(mod.q1, pred = x8, modx = x7, control.fdr = T)
```

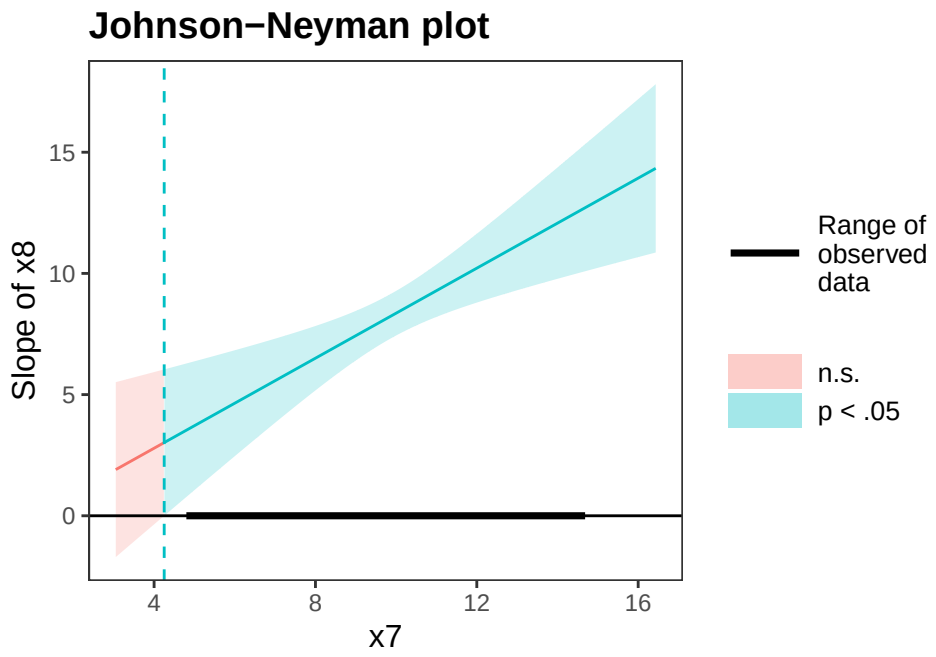
```
JOHNSON-NEYMAN INTERVAL

When x7 is OUTSIDE the interval [-9.96, 4.25], the slope of x8 is p < .05.

Note: The range of observed values of x7 is [4.89, 14.60]

Interval calculated using false discovery rate adjusted t = 1.97
```

Figure 6: Johnson Neyman interval for the model in Question 1.



Solution

the coefficient β_{11} refers to the expected change in Y for a unit increase in x_{7i} after adjusting for the simultaneous linear change in x_{8i} and vice versa.

$$\frac{\partial Y}{\partial x_{7i}} = \beta_7 + \beta_{11} x_{8i}$$
$$\frac{\partial Y}{\partial x_{8i}} = \beta_8 + \beta_{11} x_{7i}$$

Since the slope of both variables is significantly positive for the entire range of observed variables and the lines are upward sloping, the interaction is positive and increasingly strong at higher values of both variables.

1.7 Part g

Standardize the continuous data and refit the regression model. Redo part f. What do you notice about the coefficient estimates and why does this make the untangling process easier?

Solution: First, we fit the regression model using `lm()`.

```r
# Standardize Data
dat.z <- dat.q1 |>
  mutate(across(c(x1, x5, x7, x8), ~ as.vector(scale(.))))
# Refit Model
mod.q1z <- lm(y ~ x1 + x2 + x3*x4 + x5*x6 + x7*x8, data=dat.z)
```

```r
emm.q1g.z7 <- emtrends(mod.q1z, var="x7", ~x7|x8,
                       at = list(x8 = c(-1, 0 ,1))) |>
          data.frame() |>
          rename("Slope"  = "x7.trend",
```

```
                "Lower" = "lower.CL",
                "Upper" = "upper.CL") |>
        dplyr::select(x8, Slope, SE, df, Lower, Upper)
```

Table 5: Estimated marginal trends for the model in Question 1 when quantitative variables are standardized.

x8	Slope	SE	df	Lower	Upper
-1	14.57	1.24	238	12.12	17.01
0	17.86	0.92	238	16.04	19.67
1	21.15	1.36	238	18.47	23.83

```
emm.q1g.z8 <- emtrends(mod.q1z, var="x8", ~x8|x7,
                       at = list(x7 = c(-1, 0 ,1)))|>
              data.frame() |>
              rename("Slope"  = "x8.trend",
                     "Lower" = "lower.CL",
                     "Upper" = "upper.CL") |>
              dplyr::select(x7, Slope, SE, df, Lower, Upper)
```

Table 6: Estimated marginal trends for the model in Question 1 when quantitative variables are standardized.

x7	Slope	SE	df	Lower	Upper
-1	13.22	1.17	238	10.91	15.52
0	16.51	0.91	238	14.70	18.31
1	19.80	1.41	238	17.01	22.58

After standardization, a unit change represents a standard deviation change in the variable, simplifying the algebra:

- $z_{7i} = -1 \implies \beta_0 - \beta_7 + (\beta_8 - \beta_{11})z_{8i}$
- $z_{7i} = 0 \implies \beta_0 + (\beta_8)z_{8i}$
- $z_{7i} = 1 \implies \beta_0 + \beta_7 + (\beta_8 + \beta_{11})z_{8i}$

Thus, an increase of 1 s.d of x_7 corresponds to an increase of 3.29 in the slope of standardized x_8

2 Question 2

Cui and Mao (2025) investigate how freezing and cryogenic temperatures affect the strength, flexibility, stiffness, and internal breaking behavior of 3D-printed CF/PEEK composites.

Carbon fiber reinforced PEEK (CF/PEEK) is a high-performance composite used in 3D printing for aerospace, deep-sea, polar research. The researchers aimed to assess whether and how extremely cold environments impact this composite to understand how 3D-printed CF/PEEK performs in freezing conditions. This work would be important to understand its limits and develop preventative methods to protect against structural failures.

The researchers created fifteen 3D printed dog-bone-shaped test specimens and placed them in a chamber cooled with liquid nitrogen to 25°C, −25°C, −75°C, −125°C, and −175°C. To test strength of the specimen, a machine pulled both ends of the specimen until it snapped.

In the table below, the researchers record the average tensile strength for three specimens from each temperature level.

Temperature	Mean Tensile Strength
25°C	81.93 MPa
−25°C	76.63 MPa
−75°C	63.23 MPa
−125°C	32.13 MPa
−175°C	12.83 MPa

2.1 Part a

Compute the estimates we derived using calculus in lecture 16:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}_1$$
$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_{1i} - \bar{x}_1)(y_i - \bar{y})}{\sum_{i=1}^{n}(x_{1i} - \bar{x}_1)^2}.$$

Solution

$\overline{x} = -\frac{375}{5} = -75$

$\overline{y} = \frac{266.75}{5} = 53.35$

$$\hat{\beta}_1 = \frac{(25+75)(81.93-53.35)+(-25+75)(76.63-53.35)+(-75+75)(63.23-53.35)+(-125+75)(32.15-53.35)+(-175+}{(25+75)^2+(-25+75)^2+(-75+75)^2+(-125+75)^2+(-175+75)^2} \tag{1}$$

$\hat{\beta}_0 = 53.35 - .36536(-75) = 80.752$

2.2 Part b

Compute the estimates we derived using linear algebra in lecture 16:

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}.$$

Hint: Run `?matrix()` in `R` for a quick reminder about how to create a matrix.

Solution

```
X = cbind(1, c(25,-25,-75,-125,-175))
Y = matrix(c(81.93, 76.63, 63.23, 32.13, 12.83))
Beta = solve((t(X) %*% X)) %*% (t(X) %*% Y)
(Beta)
```

```
        [,1]
[1,] 80.7550
[2,]  0.3654
```

2.3 Part c

Compute the maximum likelihood estimates using `optim()` in `R` as described in lecture 16:

$$\max_{\beta_0,\beta_1,\sigma} \mathcal{L}(\theta = \{\beta_0, \beta_1, \sigma\}).$$

Solution

```
dat.q2 = tibble(
  temperatures =  c(25,-25,-75,-125,-175),
  measurements = c(81.93, 76.63, 63.23, 32.13, 12.83)
)

optim.func = function(params,data) {
  -1*sum(dnorm(data$measurements - (params[1] + (params[2] * data$temperatures)),0,sd=params[3],log=TRUE))
}

fitted = optim(
  par=c(mean(dat.q2$measurements),0,sd(dat.q2$measurements)),
  fn=optim.func,
  data=dat.q2)
(fitted$par)
```

```
[1] 80.7546760  0.3653821  6.4904683
```

2.4 Part d

Fit the model using `lm()` and plot the resulting model. Compare to the results from parts (a) through (c).

Solution

```r
mod = lm(measurements ~ temperatures, data = dat.q2)
(mod)
```

```

Call:
lm(formula = measurements ~ temperatures, data = dat.q2)

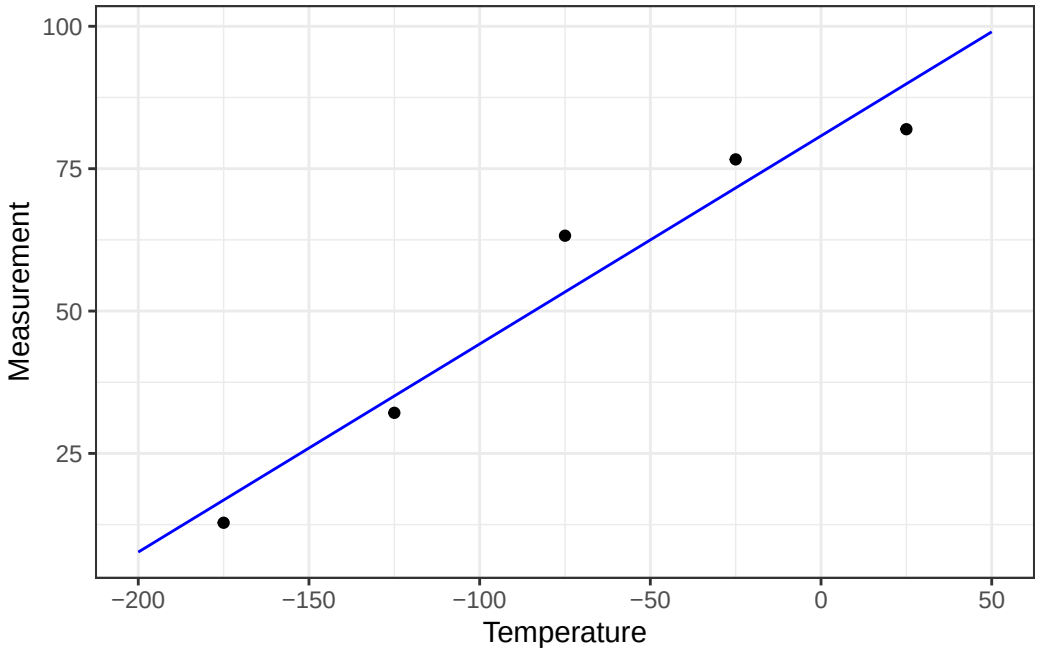
Coefficients:
 (Intercept)  temperatures
     80.7550        0.3654
```

```r
plot.dat = tibble(
  x = seq(from=-200,to=50,by=.5),
  y = coef(mod)[1] + coef(mod)[2] * x
)

plot = ggplot() +
  geom_point(data = dat.q2, aes(x = temperatures, y = measurements)) +
  geom_line(data = plot.dat, aes(x = x, y = y), color = "blue") +
  labs(x = "Temperature", y = "Measurement") +
  theme_bw()

(plot)
```

Figure 7: Fitted Linear Model of Measurement vs Temperature



The results across methods are identical, except for minor rounding differences in the by-hand calculations completed in part a.

3 Question 3

Bracht et al. (2016) evaluate MFAP4 as a biomarker for severity of fibrosis in hepatitic C patients.

```
dat.mfap4 <- read_csv("Bracht2016.csv")
```

3.1 Part a

Fit the model predicting `MFAP4` using `Fibrosis.Stage`. Write out the resulting estimate model and interpret the coefficients.

Solution

$$Y_i = \beta_0 + \beta_2 I(x_{1i} = 1) + \beta_3 I(x_{2i} = 2) + \beta_4 I(x_{3i} = 3) + \beta_5 I(x_{4i} = 4)$$

```
mod = lm(MFAP4 ~ Fibrosis.Stage, data=dat.mfap4)
(mod)
```

```
Call:
lm(formula = MFAP4 ~ Fibrosis.Stage, data = dat.mfap4)

Coefficients:
          (Intercept)  Fibrosis.StageStage 1  Fibrosis.StageStage 2
                9.230                  1.301                  3.154
Fibrosis.StageStage 3  Fibrosis.StageStage 4
               11.742                 15.388
```

Each β coefficient corresponds to the expected increase in MFAP4 level due to membership in each of the mutually exclusive groups.

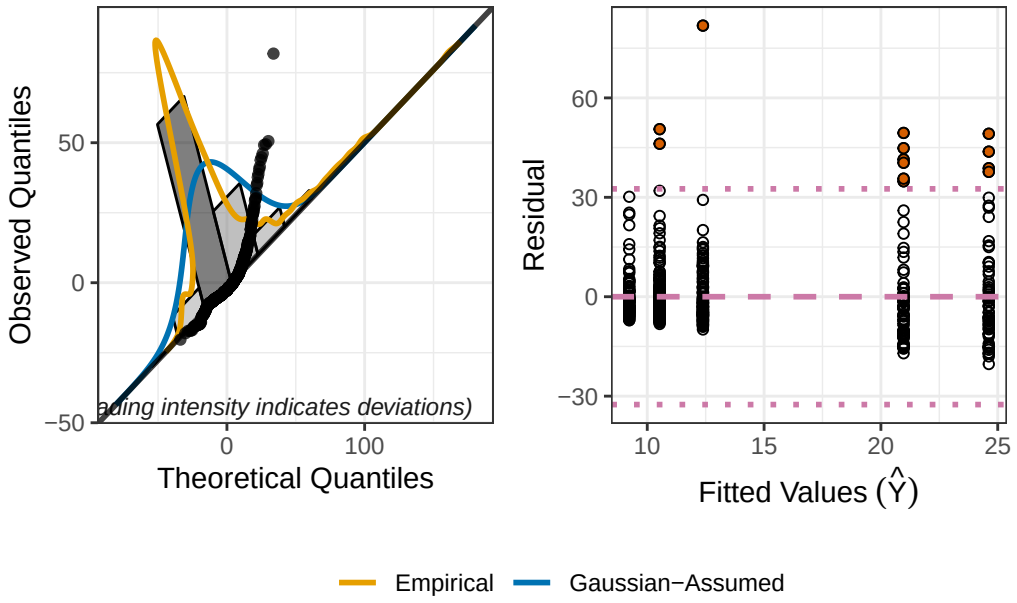
3.2 Part b

Conduct a residual analysis on the model in part a to demonstrate how to identify when you need a log transformation.

Solution

```
plotResiduals(mod)
```

Figure 8



The residuals are highly non-normal, particularly with respect to the observed quantiles deviating above the theoretical quantiles, which is consistent with right skew of residuals across most categories. The lack of normality of residuals is consistent with a need to transform the data before regressing, since this would resolve the violation of the regression condition due to the lack of residual normality and would mitigate the issue of non-constant variance of residuals.

3.3 Part c

Fit the model predicting `log(MFAP4)` using `Fibrosis.Stage`. Use the natural log, and not the $\log_2(\cdot)$ recorded in the data file. Write out the resulting estimate model and interpret the coefficients.

Solution

$$\ln(Y_i) = \beta_0 + \beta_2 I(x_{1i} = 1) + \beta_3 I(x_{2i} = 2) + \beta_4 I(x_{3i} = 3) + \beta_5 I(x_{4i} = 4)$$

```
dat.mfap4 = dat.mfap4 |>
  mutate(ln.mfap4 = log(MFAP4))

mod.ln = lm(ln.mfap4 ~ Fibrosis.Stage, data = dat.mfap4)
(mod.ln)
```

```
Call:
lm(formula = ln.mfap4 ~ Fibrosis.Stage, data = dat.mfap4)

Coefficients:
          (Intercept)  Fibrosis.StageStage 1  Fibrosis.StageStage 2
               2.0257                 0.1334                 0.3040
Fibrosis.StageStage 3  Fibrosis.StageStage 4
               0.7741                 0.9896
```

The intercept corresponds to the expected MFAP4 level in the absence of a fibrosis stage while each other coefficient corresponds to the expected increase in MFAP4 level due to membership in the associated stage.

3.4 Part d

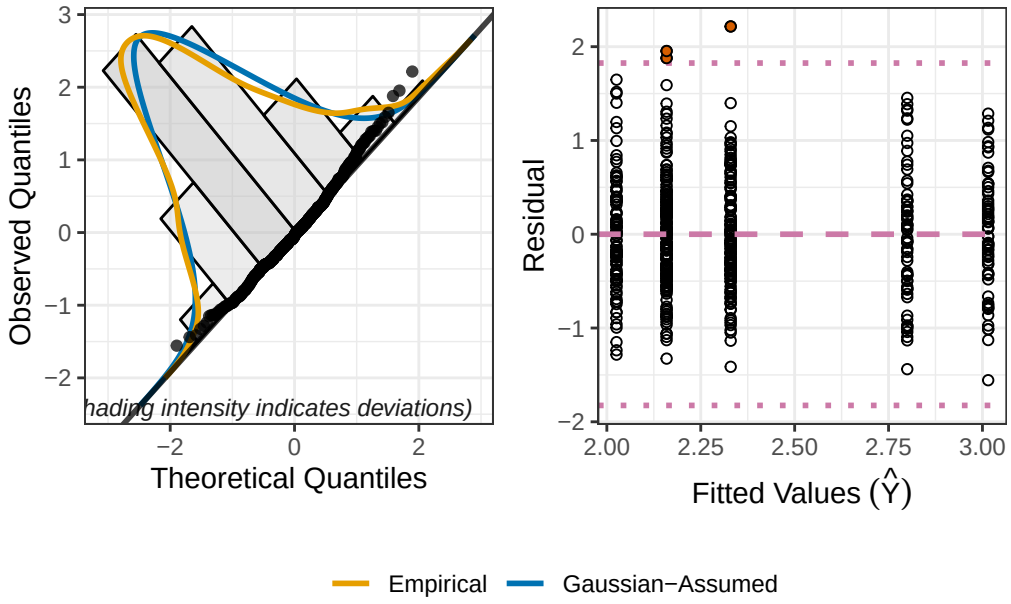
Conduct a residual analysis on the model in part c to demonstrate that the conditions for using regression are met.

Solution

$$\ln(Y_i) = \beta_0 + \beta_2 I(x_{1i} = 1) + \beta_3 I(x_{2i} = 2) + \beta_4 I(x_{3i} = 3) + \beta_5 I(x_{4i} = 4)$$

```
plotResiduals(mod.ln)
```

Figure 9



The plots show the the distribution of residuals is approximately normal, and there are very few unusually large residuals.

3.5 Part e

Fit the model predicting `log(MFAP4)` using `Fibrosis.Stage`, `Age`, and `Gender`. Use the natural log, and not the $\log_2(\cdot)$ recorded in the data file. Write out the resulting estimate model and interpret the coefficients.

Solution

$$\ln(Y_i) = \beta_0 + \beta_2 I(x_{1i} = 1) + \beta_3 I(x_{2i} = 2) + \beta_4 I(x_{3i} = 3) + \beta_5 I(x_{4i} = 4) + \beta_6 x_{5i} + \beta_7 I(x_{6i} = \text{Male})$$

```
mod.3e = lm(ln.mfap4 ~ Fibrosis.Stage + Age + Gender, data = dat.mfap4)
(mod.3e)
```

```
Call:
lm(formula = ln.mfap4 ~ Fibrosis.Stage + Age + Gender, data = dat.mfap4)

Coefficients:
          (Intercept)  Fibrosis.StageStage 1  Fibrosis.StageStage 2
              1.51867                0.12686                0.21254
Fibrosis.StageStage 3  Fibrosis.StageStage 4                    Age
```

```
   0.59887              0.82855              0.01311
Gendermale
  -0.14340
```

β_0 corresponds to the expected value of ln(mfap4) in the absence of any fibrosis stage, a female gender, and an age of 0. Each of the β coefficients $1-5$ correspond to a change membership to each of the mutually exclusive fibrosis stages from a stage of 0. the β_6 coefficient corresponds to the increase in ln(mfap4) due to a unit increase in age. β_7 is the expected difference due to the patient category changing from female to male.

4 Question 4

4.1 Part a

Simulate data to match regression assumptions in a loop that runs $R = 1000$ times.

$$
\begin{aligned}
x_i &\sim \text{Uniform}(a = 0, b = 10) \\
\epsilon_i &\sim \text{Gaussian}(\mu = 0, \sigma = 1) \\
y_i &= 2 + 1.5x_i + \epsilon_i \\
n &= 30
\end{aligned}
$$

In each iteration, fit a regression line to the data and store the coefficient estimates. Plot the resulting estimates. What do you notice?

Solution

```
regression.sims = function(r, n, error.dist) {
  beta0 = rep(0,times=r)
  beta1 = rep(0,times=r)
  for (i in 1:r) {
    dat = tibble(
      x = runif(n=n,min=0,max=10),
      e = error.dist(n),
      y = 2 + (1.5 * x) + e
    )
    mod = lm(y ~ x, data = dat)
    beta0[i] = coef(mod)[1]
    beta1[i] = coef(mod)[2]
  }

  list(
  tbl = regression.tbl(beta0,beta1),
  fig = regression.fig(beta0,beta1)
  )

}


regression.tbl = function(beta0,beta1) {
   tbl = tibble(beta0, beta1) |>
    pivot_longer(everything(), names_to = "parameter", values_to = "estimate") |>
    group_by(parameter) |>
    summarise(
      mean   = mean(estimate),
      sd     = sd(estimate),
      median = median(estimate),
      skew = e1071::skewness(estimate),
      kurtosis = e1071::kurtosis(estimate),
    ) |>
```

```
    knitr::kable(digits=2)
}


regression.fig = function(beta0,beta1) {
 pvt = tibble(beta0, beta1) |>
    pivot_longer(everything(), names_to = "parameter", values_to = "estimate")

 plt = ggplot(pvt, aes(x = estimate)) +
    geom_histogram(bins = 40, fill = "grey70", color = "white") +
    geom_vline(
      data = pvt |> group_by(parameter) |> summarise(mean = mean(estimate)),
      aes(xintercept = mean),
      color = "blue"
    ) +
    facet_wrap(~ parameter, scales = "free") +
    labs(x = "Estimate", y = "Count") +
    theme_bw()
}
```

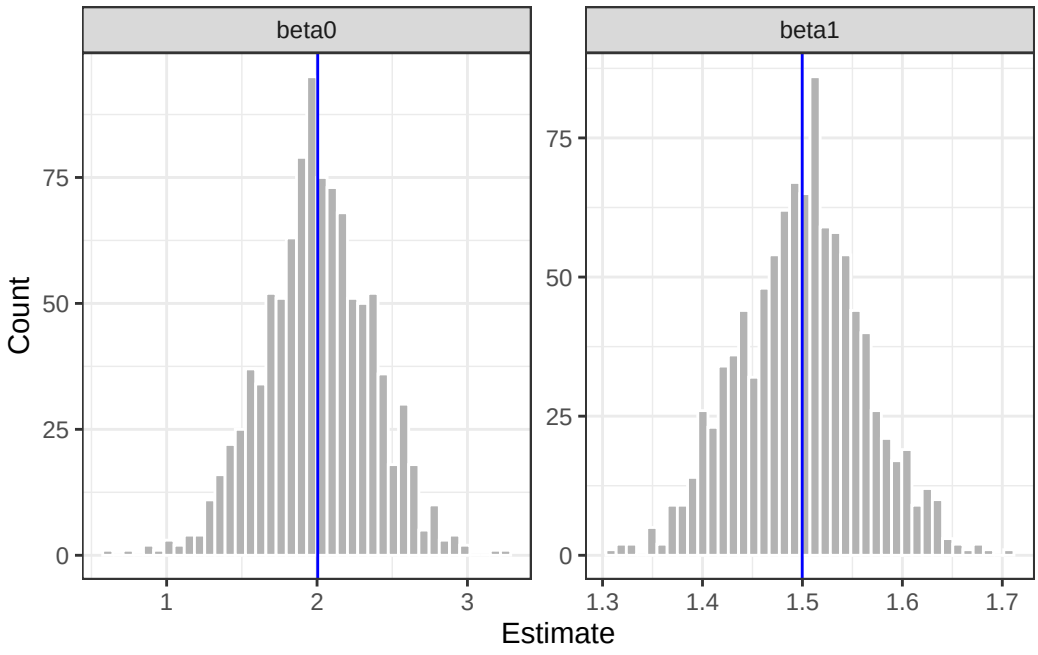
```
out = regression.sims(r=1000,n=30, error.dist= \(n) rnorm(n))
(out$tbl)
```

Table 7: Summary Statistics of Beta Coefficients with Normal Error

parameter	mean	sd	median	skew	kurtosis
beta0	2.0	0.36	2.0	-0.08	0.3
beta1	1.5	0.06	1.5	0.01	0.0

```
(out$fig)
```

Figure 10: Histogram of Beta Coefficients with Normal Error



The estimates for both coefficients are centered at the true values and normally distributed. The coefficient of

variation of β_0 appears to be noticably larger than for β_1.

4.2 Part b

Simulate data to match regression assumptions in a loop that runs $R = 1000$ times.

$$
\begin{aligned}
x_i &\sim \text{Uniform}(a = 0, b = 10) \\
\epsilon_i &\sim \text{Shifted Exponential}(\lambda = 1, \mu = 0) \\
y_i &= 2 + 1.5x_i + \epsilon_i \\
n &= 30
\end{aligned}
$$

In each iteration, fit a regression line to the data and store the coefficient estimates. Plot the resulting estimates. What do you notice?

Hint: Use the `tolerance::r2exp()` function with `rate=1` and `shift=-1`.

Solution

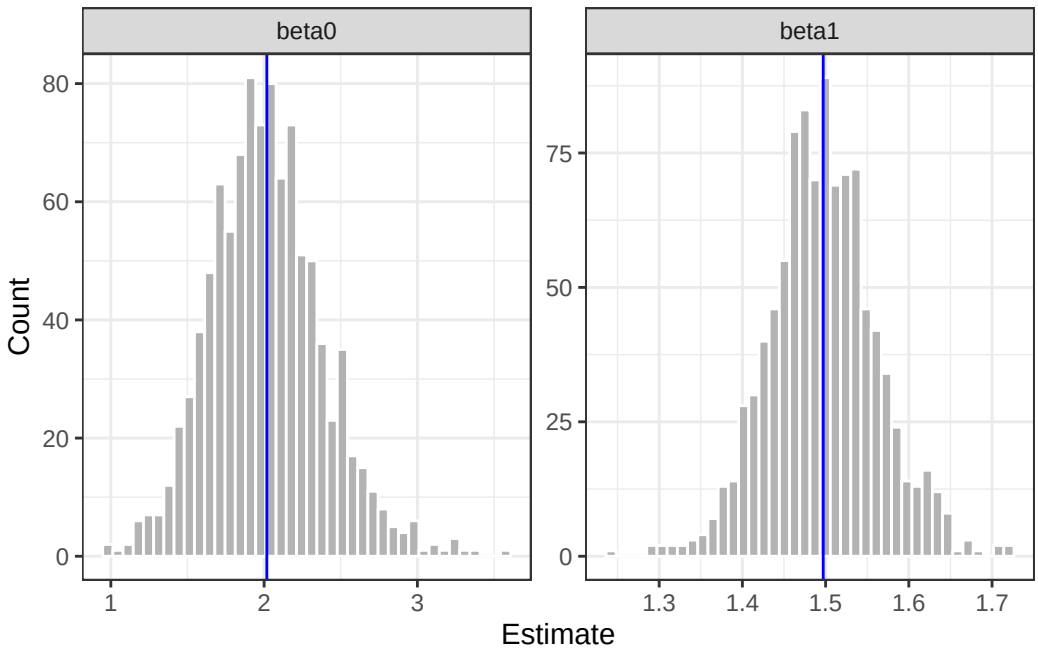
```
out = regression.sims(r=1000,n=30, error.dist= \(n) tolerance::r2exp(n,rate=1,shift=-1))
(out$tbl)
```

Table 8: Summary Statistics of Beta Coefficients with Shifted Exponential Error

parameter	mean	sd	median	skew	kurtosis
beta0	2.02	0.37	2.0	0.41	0.70
beta1	1.50	0.06	1.5	0.08	0.55

```
(out$fig)
```

Figure 11: Histogram of Beta Coefficients with Shifted Exponential Error



Both estimates are still centered at the true values, and the previous coefficient of variation observation still holds. Furthermore, the distribution of the β_0 estimate is now right skewed.

4.3 Part c

Simulate data to match regression assumptions in a loop that runs $R = 1000$ times.

$$
\begin{aligned}
x_i &\sim \text{Uniform}(a = 0, b = 10) \\
\epsilon_i &\sim \text{Cauchy}(m = 0, s = 1) \\
y_i &= 2 + 1.5x_i + \epsilon_i \\
n &= 30
\end{aligned}
$$

In each iteration, fit a regression line to the data and store the coefficient estimates. Plot the resulting estimates. What do you notice?

Solution

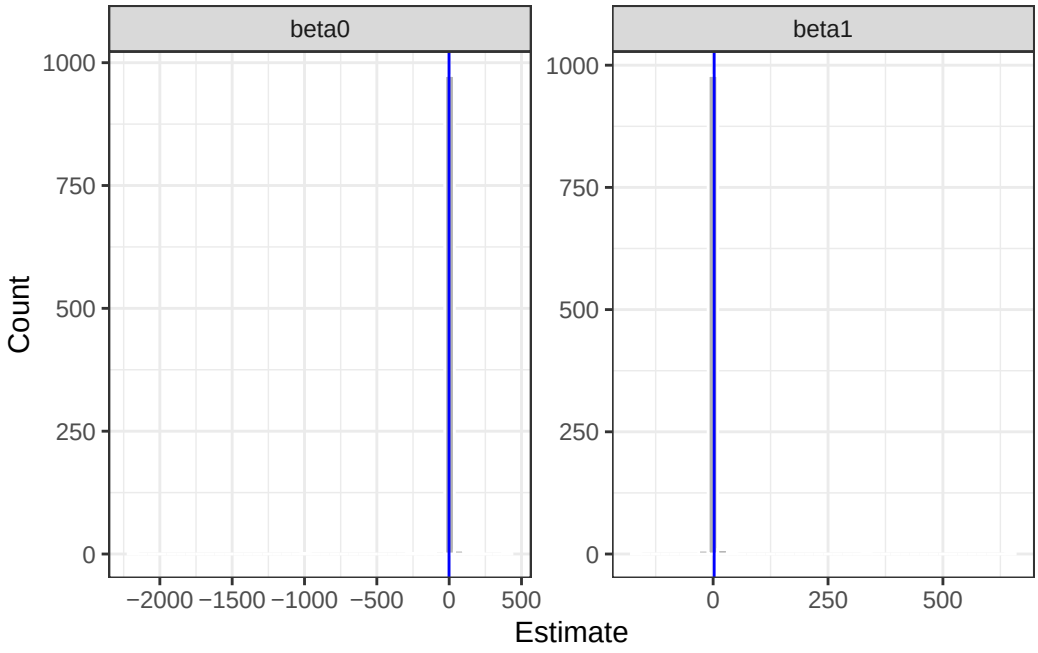
```
out = regression.sims(r=1000,n=30, error.dist= \(n) rcauchy(n))
(out$tbl)
```

Table 9: Summary Statistics of Beta Coefficients with Cauchy Error

parameter	mean	sd	median	skew	kurtosis
beta0	-1.01	79.16	2.06	-22.80	601.51
beta1	2.28	23.83	1.47	22.27	575.15

```
(out$fig)
```

Figure 12: Histogram of Beta Coefficients with Cauchy Error



The estimates are no longer centered, and the coefficient of variation and skew are now sufficiently large such that the means cannot be reasonably interpreted to be representative of the population. Also, the incredibly high kurtosis indicates fatter tails, with a greater likelihood for outliers. This would explain the distortion in the x-axis in Figure 12. Taken together, the simulation suggests standard OLS is not appropriate for models with an error term sampled from a cauchy distribution with the given parameters.

References

Bracht, Thilo, Christian Mölleken, Maike Ahrens, et al. 2016. "Evaluation of the Biomarker Candidate MFAP4 for Non-Invasive Assessment of Hepatic Fibrosis in Hepatitis C Patients." *Journal of Translational Medicine* 14 (1): 201.

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